

Artificial Intelligence (AI)

(CSCC-302)

Ch#1

What is Artificial Intelligence

A branch of computer science named Artificial intelligence pursues creating the computers or machines as intelligent as human beings.

According to the father of Artificial intelligence, John McCarthy, it is

"The science and engineering of making intelligent machines, especially intelligent computer programs"

⇒ Artificial intelligence is a way of making a computer, a computer-controlled robot, or a software think intelligently, in the similar manner the intelligent humans think.

⇒ AI is accomplished by studying how human brain thinks, and how human learn, decide, and work while trying to solve a problem, and then using the outcomes of this study as a

basis of developing intelligent software and systems.

Philosophy of AI:-

"Can a machine think and behave like humans do?"

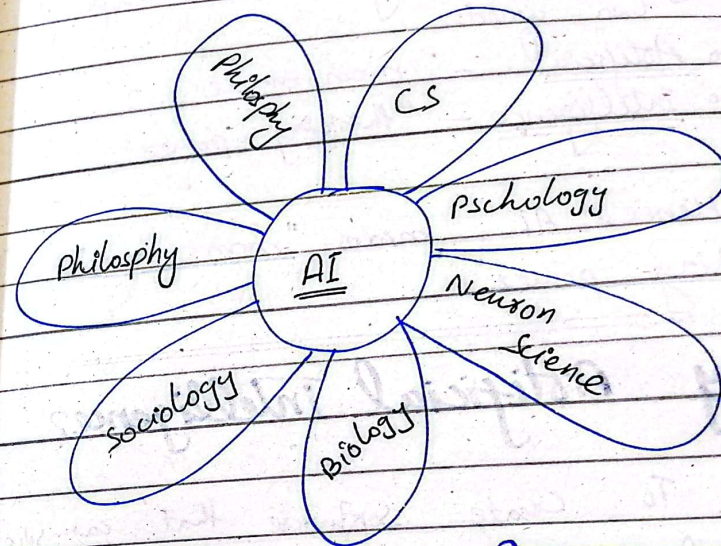
- ⇒ How does knowledge lead to actions?
- ⇒ Where does knowledge come from?
- ⇒ How does the mind arise from a physical brain?
- ⇒ Can formal rules be used to draw valid conclusions?

Goals of AI:-

⇒ Goals of AI are:

- To create Expert systems
- To implement human intelligence in machines.

What Contributes to AI?



Programming without AI

Vs

Programming with AI

⇒ A computer program without AI can answer the specific questions.

⇒ A computer program with AI can answer the generic questions.

⇒ modification leads to change the program structure and is easy not

⇒ modification can be done independently.
- Easy & Quick.

Simple definition:-

⇒ Artificial intelligence is composed of two words:

- Artificial - man-made
- Intelligence - thinking power

⇒ Hence, AI means "man made thinking power."

Why Artificial Intelligence?

⇒ To create software that can solve real-world problems.

⇒ To create personal virtual assistant such as Siri, Cortana etc.

⇒ Helps to create Robots

Goals:

- Replicate human intelligence
- Solve knowledge-intensive tasks

Pros of AI:

- High Accuracy will less errors
- High Speed
- High reliability
- Digital Assistant

Cons of AI:

- High Cost
- Can't think out of the box
- No feelings & emotions
- No original Creativity

Prerequisite of AI:

- C, C++, Java, Python etc.
- Knowledge of essential mathematics such as derivatives & probability theory etc.

Applications of AI:

- Gaming
- Natural language processing
- Expert Systems
- Vision Systems
- Speech recognition

- Handwriting Recognition
- Intelligent robots

The Foundations of AI

⇒ The foundation provides the disciplines that contributed ideas, viewpoints and techniques to AI.

- Philosophy
- Mathematics & Statistics
- Economics
- Neuroscience
- Psychology
- CS & Engineering
- Control theory & Cybernetics
- Linguistics

(1) Philosophy:-

⇒ Philosophy is the very basic foundation of AI.

⇒ "The study of fundamental nature of knowledge, reality and existence are considered for solving a specific problem is a basic thing in artificial intelligence."

⇒ Philosophy defines:
"How can the formal rules be used to draw valid conclusions."

⇒ With out philosophy it is difficult to answer the following questions.

- How does the mind arise from a physical brain?
- Where does the knowledge come from?
- How does the knowledge lead to action?

(2) Mathematics & Statistics

— provides framework for models

⇒ AI services formal Logic & probability for planning and learning.

⇒ Computation services for analyzing relation, & implementation.

⇒ Knowledge in formal representation are most services for writing actions for agents.

⇒ In AI, the mathematics & statistics are most important for

- Proving theorems
- Writing algorithms
- Computation
- Decidability
- Tractability
- modeling - Uncertainty
- Learning from data

⇒ What are the formal rules to draw valid conclusions?

⇒ What can be computed?

⇒ How do we reason with uncertain information?

(3) Economics

⇒ Deals with investing the amount of money, and maximization of utility with minimal investment.

⇒ While developing an AI product, we should make decisions for

- when to invest?
- How to invest?
- How much to invest?

o where to invest?
⇒ To answer these questions one should have knowledge about Decision Theory, Game Theory, Operations Theory & etc.

(4) Neuroscience:

⇒ Neuroscience: is the study of the nervous system, particularly the brain.

⇒ Human brains are somehow different, when compared to other animals. Man has the largest brain in proportion to his size.

⇒ The brain consists largely of nerve cells or neurons. The observation of individual neurons can lead to thought, action, consciousness of one's brain?
⇒ How do brains process information?

(5) Psychology / Cognitive Science

The scientific method to the study of human vision.

- o Problem Solving Skills
- o How do people behave?
- o Perceive
- o Process Cognitive information, and represent knowledge.
- o How do humans and animals think and act?

(6) Computer Science & Engineering:-

Logic and inference theory, algorithms, programming languages and system are important parts of Computer Science.

Computer hardware gradually changed for AI Applications such as GPU, TPU and wafers scale engine (WSE).

⇒ The amount of computing power used to train top machine learning applications and the utilization doubled every 100 days.

⇒ The super computers & quantum computers can solve very complicated AI problems.

(7) Control Theory

⇒ Control theory helps the system to analyze, define, debug and fix errors by itself.

⇒ Developing:

- Self-controlling machine
- Self-regulating feedback control systems
- The submarine

are some examples of control theory.

⇒ Calculus and matrix algebra and the tools of control theory provide themselves to

systems that are describable by fixed sets of continuous variables, are foundation of AI.

⇒ Knowledge representation, grammars, computational linguistics or natural language processing are significant to developing AI Applications.

⇒ The tools of logical inference & computation provide the language, vision, and symbolic planning of agent programming.

◦ How can artifacts operate under their own control?

(8) Linguistics

⇒ Speech recognition is a technology which enables a machine to understand the spoken language and translate into a machine readable format.

⇒ It is a way to talk with a computer, and on the basis of that command, a computer can perform a specific task.

⇒ It includes Speech to Text, Text to speech.

◦ How does language relate to thought?

History of Artificial Intelligence

1941: The initial computer, even to run a single program they need to do many connections and it is used to be complex task to do.

1943:- First work recognized by warren mcculloch and walter pits proposed a model of AI Neurons.

1949: Donald Hebb → update and modify connection strength b/w neurons called Hebbian learning.

1950:- Alan Turing proposed a test "Computing machinery & intelligence" Test to check machine ability by human intelligence.

1956: Birth of AI - Darts month conference adopt was artificial intelligence by American Scientist (John McCarthy).

1966: Joseph weizenbaum → first chat box - Eliza

1972: first intelligent humanoid robot - WABOT-1 made japan.
- (1980 - AI wines)

1980: AI come back with expert system, they are programmed with decision making abilities
- (1993 - AI wines and)

1997: IBM deep blue beat world chess champion.

2002:- AI entered in the home in form of Roomba a vacuum cleaner

2006: AI in business world, twitter etc

2011: IBM watson

2012: Google launch Android app

2014: Chatbot "eugene grossman"

Q. What is Total-Turing tests

⇒ Coined in 1950 a test by Alan Turing
"Turing test is used to determine whether or not machines can think intelligently like humans".

⇒ The Turing test is a way to check if a machine can have a conversation that's so human-like, a person cannot tell if they are talking to a machine or a human. If the machine fools the person, it's considered to have passed the test.

⇒ In the test, a human judge interacts with both a machine and a human through a text-based interface, and if the judge cannot reliably distinguish which is the machine & which is the human based on their responses, the machine is said to have passed the Turing test.

According to Total Turing test:-

The total Turing Test, also known as the Complete Turing Test, is an extended version of the original Turing test.

In addition to text-based conversation, it includes other modes of communication, such as audio and video, to assess a machine's ability to exhibit human-like intelligence across different forms of interaction. Passing the total Turing Test means the machine can imitate human behavior not only in text-based conversation.

Q. What a computer needs to possess if a computer is really intelligent?

→ The computer needs to possess the following capabilities:

- Natural language processing
- Knowledge representation
- Automated reasoning
- Machine learning